CubeSats for microbiology and astrobiology research

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1 Introduction

The study of microbiology in outer space falls within two broad categories linking biological science to space: astrobiology and fundamental space biology. Astrobiology seeks to satisfy human curiosity by expanding scientific understanding of life's origins, evolution, distribution, and future in the universe, including such topics as prebiotic chemistry, extraterrestrial adaptation and survival, and searching for habitable environments and nonterrestrial life. Fundamental space biology focuses pragmatically on the potential for terrestrial life to adapt and survive away from Earth. Critical studies are conducted by carrying organisms into space and studying every aspect of how that environment affects them. Space's primary perturbations to life are diminished gravitation and a complex, variable spectrum of ionizing radiation; studies of adaptation and basic survival are augmented by characterizing the degree to which organisms, communities, or ecologies can thrive and maintain functional performance relative to reference terrestrial conditions.

Both astrobiology and fundamental space biology are ripe for scientific advances achievable with small, microbial payload experiments that can be carried aboard CubeSats, albeit with limitations. For example, in astrobiology neither the study of prebiotic chemistry nor the search for extraterrestrial life requires carrying microbes into space, and while the study of microbes is central to fundamental space biology, complex multicellular organisms—up to and including humans—are also of great interest but not CubeSat appropriate.

Despite such limitations, emphasis by the world's large space agencies on the complex, costly task of supporting humans in space has led to a variety of basic biological studies utilizing so-called model organisms, many of which are microbes and thus generally CubeSat compatible. These microbiology experiments aim to anticipate and explain space environment effects on humans: the model organisms are well and widely characterized in terrestrial labs, and their similarities to, and overlaps with, cells and functional processes in the human body are understood and exploited.

Indeed, model organisms were the subjects of the first life science space studies conducted aboard early Soviet and American missions: In 1960 Korabl-Sputnik 2 carried *Escherichia coli*, and Discoverer 17 ferried *Clostridium sporogenes* into space [1–3]. Subsequently, the Soviet Vostok spacecraft, the Mir space station, and many

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of the US Gemini, Apollo, and Space Shuttle missions supported space microbiology experiments. Basic biological studies were never, however, the core reason for these missions. This is now changing, thanks to CubeSats' enabling spaceflight missions designed explicitly to conduct microbiology experiments.

Given their academic origins, the first CubeSat developments and missions focused understandably on technology demonstration and student training—the earliest CubeSat science endeavors included physical characterization of the space environment and observations of Earth [4,5]. In 2006, this scientific scope expanded measurably when the *GeneSat-1* mission demonstrated that CubeSats can serve as Earth orbit science platforms for living microbes, thus providing the first opportunities for microbial biological studies to be the principal focus of entire spaceflight payloads [6].

From an engineering point of view, the establishment of a bus that permits interchangeable or even standardized payloads can streamline and hasten microbiological space research capability development, which in turn stands to reduce programmatic costs. This chapter presents the results and future potential of doing microbiology experiments on a CubeSat platform, including the benefits conferred to space biology research.

To date, NASA's Ames Research Center is the only institution to have operated nanosatellites with live biological payloads. The chronological listing in Table 1 reveals NASA Ames' evolutionary approach to the development of the five space-life-sciences CubeSats flown to date by using the technologies of each satellite and its

Table 1 Biological or astrobiological CubeSat missions.

Mission (format)	Description	Launch year	Outcome
GeneSat-1 (3U)	• 2U payload measured expression of green fluorescent protein in *Escherichia coli* and tracked microbe population via light scattering • First NASA nanosatellite mission; first biological payload to fly in space on a CubeSat platform	2006	Full mission success
PharmaSat (3U)	• 2U payload measured antifungal drug dose-response curves for *Saccharomyces cerevisiae* fungus using colorimetry to measure metabolic	2009	Full mission success

Table 1 Continued

Mission (format)	Description	Launch year	Outcome
	activity and population versus time • First NASA principal investigator-led nanosatellite mission		
O/OREOS[a] (3U)	• Two independent 1U astrobiology payload modules measured (i) survival of *Bacillus subtilis* up to 6 months and (ii) long-term photodegradation of biomarkers and bio-building blocks for 1.5 years via UV-visible spectroscopy • High-radiation, high-inclination orbit; deorbit mechanism	2010	Full mission success, both payloads. Spacecraft remained operable ~5 years in orbit
SporeSat-1 (3U)	• 2U payload to measure gravitational response of *Ceratopteris richardii* fern spores via Ca^{2+} ion channel response • Variable gravity in microgravity ambient using 50-mm microcentrifuges with 32 ion-specific [Ca^{2+}] electrode pairs	2014	Successful space demo of mini centrifuges with integral ion-sensitive electrodes
EcAMSat[b] (6U)	• 2U payload measured antibiotic resistance in microgravity versus dose for uropathogenic *Escherichia coli* • 6U format provided 50% more solar-panel power to keep payload experiment at 37°C for extended durations	2017	Full mission success

[a] *O/OREOS* = Organism/Organic Exposure to Orbital Stress.
[b] *EcAMSat* = *Escherichia coli* Antimicrobial Satellite.

payload as starting points and stepping stones for the next. This approach reduces risk and cost while increasing the probability of mission success.

GeneSat-1 was designed, built, and flown to demonstrate that in situ biological research including integral microbial life support in a CubeSat was feasible without return of any biological samples [7]. *PharmaSat*, the first nanosatellite to host a competitively selected, peer-reviewed, science-driven mission, improved upon *GeneSat-1*'s 1U bus and took multiple lessons from its payload technologies [8]. The bus of the Organism/Organic Exposure to Orbital Stresses (*O/OREOS*) satellite was a modestly improved version of the 1U bus of *PharmaSat* [9–11]; this mission was the first dedicated to astrobiology and the first to support two completely independent 1U bioscience payloads. *EcAMSat* (*E. coli* Antimicrobial Satellite), selected by peer review from an open public opportunity, was the first 6U microbiology CubeSat; it not only utilized a copy of the 1U *PharmaSat* bus but also repurposed over 90% of *PharmaSat's* payload system as well. *SporeSat*, selected for development during the same public competition as *EcAMSat*, utilized a new 1U bus design featuring a real-time operating system; its 2U payload implemented multiple new technologies, including variable-rate microcentrifuges with integral illumination and a pair of ion-sensitive electrodes for every specimen. The university-operated ground station and mission operations approach evolved from *GeneSat-1* to *PharmaSat*, remaining essentially the same for *O/OREOS* and the next missions to fly [8,12,13].

This chapter surveys and summarizes progress to date with "microbiology CubeSats" from the science, engineering, and programmatic points of view. In addition to their science payloads, key satellite "bus" subsystems, for example, power and communication, and their ground control systems are described. Operational parameters such as orbital information are also detailed. Special focus is given to the payload hardware design, development, and operation. Finally, the first biology experiment to be conducted in heliocentric orbit beyond Earth's magnetosphere since the final Apollo mission in 1972 is described and discussed: *BioSentinel* is a CubeSat scheduled for launch as soon as 2021.

2 CubeSats for microbiology research

This section will present all the CubeSats missions carrying microbiological experiments mentioned above. In all cases, the core of the payload was one or more microfluidic cards or disks, with multiple wells, placed inside a hermetic container filled with humidified air at 1 atm to allow for aerobic respiration [7,14–16].

2.1 GeneSat-1

GeneSat-1's scientific objective was to characterize bacterial growth and metabolics using *E. coli* as the model organism. This implied growing bacteria in a suitable controlled environment and acquiring and transmitting data back to the scientists. These experiment objectives created the following mission requirements, constraints, and drivers [7]: (i) a microfluidic-based payload needed to be capable of operations in microgravity, (ii) the payload temperature needed to be regulated to within ±0.5°C of a defined set point, and (iii) high-quality optical sensors needed to be miniaturized for in situ data acquisition, preferably without moving parts. The core of *GeneSat-1*'s payload was a 12-well (110 μL/well) card, including two wells as solid-state controls. The card was an acrylic manifold that connected the 10 wells to an input and an output port; each well had a pair of 0.22-μm nylon fiber membrane filters at its inlet and outlet to keep *E. coli* in the wells during fluid exchange. One of the faces of the card was covered with a 0.5-mm-thick optical-quality acrylic plate and the other with a 75-μm-thick gas-permeable membrane (polystyrene). The card was connected to a 15-mL polyethylene vinyl acetate (PEVA) bag filled with growth medium through a solenoid valve (Parker). Similarly a second PEVA bag was connected to the card outlet. Each of the 12 wells had a 3-W blue LED to produce fluorescence excitation (470 nm) and a 2.3-mW green LED for illumination to acquire optical density. Data were acquired with an intensity-to-frequency detector and a 525-nm emission filter to detect green fluorescence as a proxy of gene expression and to pass the green LED wavelength for light scattering to quantify cell counts [6]. The major payload components are shown during the integration process in Fig. 1.

As a secondary payload, a late load (or reload) was not possible for *GeneSat-1*. The reagents and biological samples had to survive a minimum of 6 weeks before launch, including a 4-week prelaunch period plus 2 weeks in case of launch delays—the time to operation in space ended up being just under 7 weeks due to launch slips [17]. Prior to payload integration, cells were loaded in the microfluidic card wells in a PBS-based stasis buffer to place them in metabolic dormancy. Two strains of K-12 *E. coli* were loaded into the fluidic cards for flight and for a parallel ground control experiment: DH5alpha (Arizona State University), with plasmid AcGFP to express green

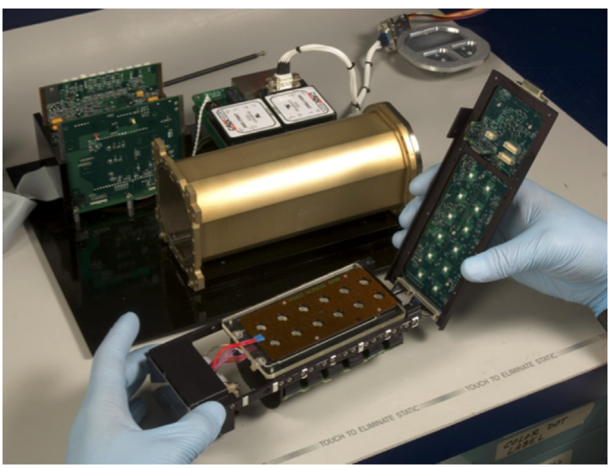

Fig. 1 *GeneSat-1* payload components during integration. Fluidic card (sandwiched between heater plates), in *left hand*, with blue excitation and combined detection system beneath it; circuit board with green LEDs in *right hand*; hermetic container in the background. Credit: NASA.

fluorescent protein (GFP), and MM294 cells (Carolina Biological), with plasmid p-GREEN for GFP. Both engineered organisms express GFP constitutively (emission peak: 512 nm).

The mission launched into a 40-degree inclination, 413-km perigee, and 424-km apogee orbit, on December 16, 2006, aboard a Minotaur I rocket from the Wallops Flight Facility in Virginia, USA. Once the experiment was initiated, the card was warmed to 34°C, and growth medium was introduced into the wells, replacing the stasis buffer and initiating cell growth. Fluorescence and cell growth were monitored using the optical detection system described earlier. During flight, communication was limited to a few passes per day through prearranged ground stations, limiting the possibility of human control. Due to these limitations, *GeneSat-1* was designed for fully autonomous operations, including initiation of the experiment, temperature control, biological measurements, data recording, and telemetry.

GeneSat-1 was a fully successful mission; fluorescence and cell growth were observed in all nine microfluidic wells containing one strain or the other of *E. coli*; one well carried no microbes and produced a flat baseline throughout the experiment, as did the solid-state controls. During the exponential phase of growth, the two strains of *E. coli* doubled with an average period of ~50 min per generation in space microgravity and ~35 min per generation on Earth. The slower growth rate in space was tentatively attributed to the absence of gravity-driven forces and flows in microgravity, namely, thermal convection, buoyancy, and sedimentation, which means that delivery of nutrients and removal of waste products to/from the cells occur only by diffusion. This altered extracellular mass transport phenomenon has been corroborated via a molecular genetic analysis [18]. The spaceflight results demonstrated a system capable of autonomous operations in low Earth orbit [17].

2.2 PharmaSat

PharmaSat's scientific mission objective was to study the effects of microgravity on yeast (*Saccharomyces cerevisiae*) growth and metabolism and on antifungal drug efficacy via three-color optical absorbance. To achieve these goals, the satellite needed to accomplish four main functions: (i) provide life support for the microbes in the fluidic card in the payload, (ii) introduce antifungal agents into the wells at several concentrations, (iii) measure optical density in the wells to calculate population growth, and (iv) measure culture viability via a metabolic indicator dye [8,16,19]. The payload hardware was contained within a 1.2-L hermetically sealed vessel [16], which in turn was covered in multilayer insulation (MLI) blankets and gold plating for thermal optimization [19]. To reduce heat conduction, the payload was attached to the bus with titanium screws and Ultem washers. The payload components were warmed with a pair of flexible patterned nichrome-on-Kapton 2 W heaters (Minco), each bonded to an aluminum thermal spreader plate for spatial and temporal thermal uniformity [19]. The heaters were capable of providing a temperature stability of better than ±0.3°C.

The 6.4 × 12.8 cm fluidics card (the same length as standard 96-well plates but narrower due to payload size limitations), Fig. 2, was made of laser-cut poly (methylmethacrylate) layers and included 48,100-μL wells (4-mm diameter and 7.8 mm deep,

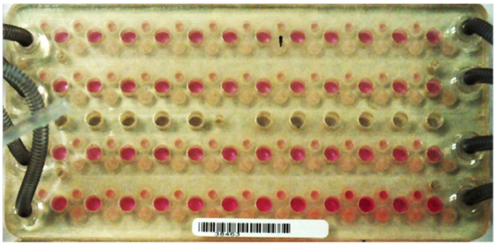

Fig. 2 *PharmaSat* fluidic card with four banks of 12 fluidic wells designated, from top to bottom, for high, medium, low, and zero dosage of antifungal drug. The *pink color* (light gray in print version) is due to alamarBlue that has been converted by metabolites of growing yeast to its *pink* (reduced) form; *pink* is most intense in the control bank, and faintest in the high-dose bank, due to relative rates of metabolic activity. Center bank of dry wells was reserved for solid-state color standards.
Credit: NASA.

spaced 9 mm center to center) and 11 solid-state reference wells. Every fluidic well had 1.2-μm nylon fiber membrane filters at its inlet and outlet to prevent removal of yeast from the wells during fluid exchanges. The card included four independent manifolds serving the four banks of wells to support three separate drug dose levels plus a zero-dose control. There was a 51-μm-thick polystyrene gas-permeable membrane on each side of the fluidic card [16]. The payload hardware also included a miniaturized environmental control system, microfluidics system with pumps, valves, and an array of 59 optical sensors, one per well [8].

PharmaSat launched as a secondary payload aboard a Minotaur I rocket from Wallops Flight Facility on May 19, 2009. As with *GeneSat-1*, no late load was allowed, and no power was available for thermal control prior to deployment in space. During payload assembly, yeast cells were transferred into stasis buffer and loaded into the fluidic card microwells. After launch, *PharmaSat* was spring ejected into low Earth orbit (40-degree inclination, 432-km perigee, and 467-km apogee). The experiment was initiated by feeding the cells with growth medium containing a metabolic indicator dye, thus displacing the stasis buffer. Each microwell within the fluidic card was monitored using a three-LED optical source and detector system. This system measured yeast growth in two ways: (i) via optical density changes due to light scattering by the yeast cells, which is directly proportional to cell number, and (ii) via color change of the viability dye, alamarBlue. This dye changes from its oxidized blue form to its reduced pink form when the cells become metabolically active, at a rate dependent on the number of cells and their average metabolic output. After the cells were allowed to recover from stasis for 12 h (and before reaching high cell density), the fluidic system introduced three different doses of the antifungal agent voriconazole [16]. This antifungal acts by disrupting the fungal cellular membrane. Cell growth and viability after antifungal exposure were tracked with measurements every 15 min for 72 h. An identical satellite system housed in a thermal chamber was used as a delayed synchronous ground control unit to compare ground and flight results [16].

Comparison of zero-drug-dose results from spaceflight microgravity with results from the ground experiment showed that spaceflight yeast had a longer "lag time" than the corresponding ground specimens, whether measured by cell density (via turbidity) or metabolism (via alamarBlue depletion). Also, spaceflight samples exhibited a slower growth rate than those on the ground, again regardless of whether this was measured by metabolism or population doubling time during the exponential growth phase. At low and medium voriconazole concentrations, *once the difference in rates of the zero-dose controls was taken into account by normalization*, there was no significant difference between spaceflight and ground specimens in the rates of either cell growth or metabolic activity. *PharmaSat* was the first fully autonomous pharmaceutical dose-response system on a free-flying satellite.

2.3 SporeSat

SporeSat was developed to investigate the effect of gravity on the reproductive spores of the aquatic fern *Ceratopteris richardii*. Given its goal to improve understanding of biological gravity sensing, *SporeSat*'s objective was to measure calcium concentration that results from the opening and closing of calcium ion channels in these spores using lab-on-a-chip devices called biology compact discs (bioCDs), which allow for real-time measurement of calcium signaling using differential pairs of ion-sensitive electrodes, while the discs are rotated to create artificial gravity. To differentiate the role of the gravitational regime from other aspects of the spaceflight environment (e.g., radiation), *SporeSat* adapted the miniature centrifuge developed initially by the *GraviSat* project [20], integrating on-disk ion-selective microelectrodes, as well as LED illumination, to initiate and monitor the gravitational dependence of the germination of fern spores. Two such 50-mm bioCDs were to be rotated at defined rates to simulate variable gravity in a series of steps from 0.06 to 2 g, while a third one remained stationary as a microgravity control [21]. This created new requirements, constraints, and drivers on the satellite design and implementation. The novel sensor discs flew in a 3U nanosatellite that utilized flight-proven technologies previously flown in *GeneSat-1*, *PharmaSat*, and *O/OREOS*.

Fern spores and media were loaded onto three bioCDs (32 spores each; each spore had its own 160-μm-diameter well with integral ion-sensitive electrodes), then integrated into the CubeSat; see Fig. 3. *SporeSat-1* launched on April 18, 2014, on a SpaceX Falcon 9 rocket as a secondary payload on the International Space Station (ISS) resupply mission CRS-3. It was deployed en route to ISS at an altitude of 325 km and an inclination of 51.6 degrees; it reentered Earth's atmosphere after 47 days. The experiment was initiated by increasing temperature to 29°C and establishing artificial gravity (bioCD-1 and bioCD-2). While the experimental plan called for initiation of germination via red light spore activation for a duration of several hours, the large-area red organic light-emitting diode (OLED) failed to illuminate in the space experiment (with a similar failure for the ground control). Despite this, the rotating bioCDs were held at the desired temperature, the rotation occurred at the defined rates, and differential calcium ion signals were measured from each one of the 96 spores, albeit at nominal background levels expected in absence of biological

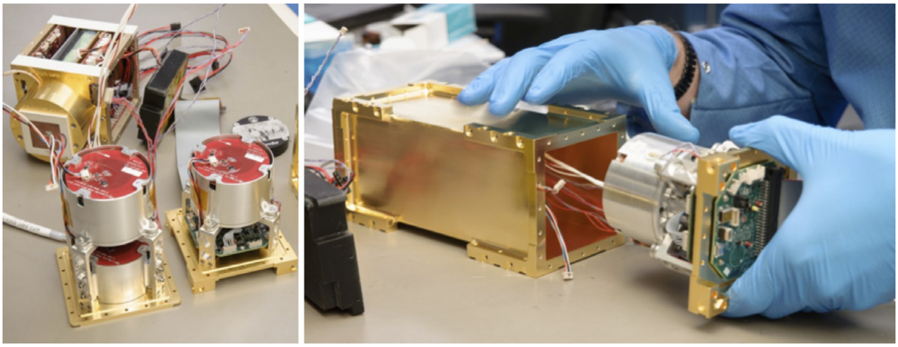

Fig. 3 *SporeSat*'s two mini centrifuges with integral bioCDs (*left*, foreground) and bus (background); the left-hand centrifuge sits over an identical nonrotating bioCD (bioCD-3). *Right*: inserting one mini centrifuge into the payload hermetic container.
Credit: NASA.

activity, with noise levels in the 10s of microvolts. Thus, *SporeSat-1* demonstrated several technologies relevant to measurement of spore gravitational sensing in a 3U CubeSat format, including the capability to generate artificial gravity while measuring ion channel response on a rotating bioCD [21].

SporeSat-2 was developed to address software issues from *SporeSat-1* and to prepare a more robust means of illuminating fern spores to trigger their germination, namely, a set of 32 individual surface mount LEDs per bioCD, one directly opposite each spore-containing microwell. *SporeSat-2* was extensively ground tested [21] but is not currently scheduled to fly as a space mission.

2.4 EcAMSat

EcAMSat's mission was to investigate the effects of microgravity on the dose-dependent antibiotic response and resistance in uropathogenic *E. coli* [22] and to do so by heavy reuse of *PharmaSat* designs and spare hardware. *EcAMSat* was the first 6U biological CubeSat and the first biological CubeSat to be deployed from ISS. Two strains of *E. coli*, a wild type and a mutant strain from which the stress-relevant *rpoS* gene had been deleted, were inoculated in 48 independent microwells in a fluidic card in stasis buffer. *E. coli* cells remained in stasis for approximately 8 weeks before cell growth was initiated. *EcAMSat* launched aboard the Orbital ATK OA-8 resupply mission to the ISS on November 12, 2017, and deployed from the Japanese Experiment Module (JEM) on November 20 (Fig. 4) (ISS' orbit is 51.6 degrees, 408-km perigee, and 410-km apogee). Once in space, growth medium was delivered into the wells, and the cells were allowed to grow to stationary phase; growth was monitored via optical density. After a starvation period, bacteria were exposed to three dose levels of the antibiotic gentamicin (12 wells per dosage and 6 for each strain) along with 12 wells of zero-dose controls. This antibiotic belongs to the aminoglycoside class of antibiotics, which inhibits bacterial ability to synthesize

Fig. 4 Photograph of the 6U, 11-kg *EcAMSat* nanosatellite *(lower left-center)* above Earth seconds after deployment from the ISS (deployer is at *mid-upper center* of photo). It was the first 6U biology satellite and the first biological science satellite of any size to be deployed from ISS. Credit: NASA.

proteins by binding to the 30S subunit of the bacterial ribosome. The antibiotic was then replaced by a solution of the metabolic indicator alamarBlue, and bacterial metabolism was measured by the rate of reduction of the dye. Optical measurements were performed using the same three-color LED detection system (470, 525, and 615 nm) used for *PharmaSat,* with measurements at 15-min intervals throughout the experiment. Heaters with embedded temperature sensors were attached to the fluidic card to allow for temperature control and ensure cell survival. Control experiments were performed in parallel several days later on the ground. The results of the development and ground testing of the *EcAMSat* payload are reported elsewhere [22], and the science results of the *EcAMSat* spaceflight experiment have been published [23].

3 CubeSats for astrobiology research

Thus far, only one CubeSat mission has had an astrobiology research focus: *O/OREOS*. This CubeSat carried two payloads, Space Environment Survivability of Life Organisms (SESLO) and Space Environment Viability of Organics (SEVO), each with its own astrobiology mission objectives [12]. SEVO characterized four organic compounds, an amino acid, a quinone, a polycyclic aromatic hydrocarbon (PAH), and a metalloporphyrin, as they were exposed to the space environment [9–11], particularly full-spectrum solar illumination extending deep into the ultraviolet (124 nm). The objective of SESLO was to measure long-term survival, germination, and metabolic activity of *Bacillus subtilis* spores exposed to microgravity and ionizing radiation for up to 6 months [14]. The *O/OREOS* spacecraft bus and mechanical configuration, as well as many aspects of the SESLO payload, were derived from the *GeneSat-1* and *PharmaSat* 3U nanosatellites. Besides exploring the microgravitational regime, *O/OREOS* included radiation as another independent variable because of its highly inclined orbit and altitude (72 degrees, 621-km perigee, and 646-km apogee), which made it travel through relatively weak regions of the magnetosphere twice per 98-min orbit: the estimated radiation dose rate was about 15 times that typical of an ISS-like orbit [12]. The SESLO payload, Fig. 5, was 1U in size and had three "bioblocks." Each block has 12 75-μL wells interconnected through microfluidic channels and a solenoid valve to a reservoir with germination medium colored with alamarBlue. Each block was used to independently assess growth at three different times: 14, 97, and 180 days after launch. Three different wavelengths were used to acquire data: 470, 525, and 615 nm. As alamarBlue reacted with metabolites in the cells in microwells, it changed from blue to pink and then from pink to colorless [14].

The *O*/*OREOS* nanosatellite was launched as a secondary payload on November 19, 2010, aboard a Minotaur IV rocket from Kodiak Launch Complex, Alaska. In SESLO, bacterium spores of two different strains of *B. subtilis* were first loaded and dried onto the walls of the bioblock microwells and maintained in desiccated state prior to growth. Fourteen days after launch the first SESLO bioblock was activated. After growth, target temperature stabilized at 37°C; then growth medium was delivered into the microwells, allowing rehydration of the spores. This was the first CubeSat experiment to fly its biological specimens in the dried state and rehydrate them in space. Germination, growth, and metabolism were monitored for 48 h. Optical readings were performed at 15-min intervals throughout the experiment. Delayed-synchronous ground control experiments were performed with identical hardware with biological samples loaded from the same flight cultures.

Very similar results were recorded at the 14- and 97-day timepoints from SESLO in space and its ground control. The main difference anticipated between the space microgravity and 1-g control environments was the previously explained altered mass transport phenomenon [18]. Thus it was not surprising that the blue-to-pink color transitions of alamarBlue for the ground controls occurred on average about 40 min sooner than they did for the spaceflight microwells. The other notable difference was between the 168 wild-type *B. subtilis* and the mutant strain, WN1087: the latter strain clearly metabolized alamarBlue more rapidly than the former, both on the ground and in space [14].

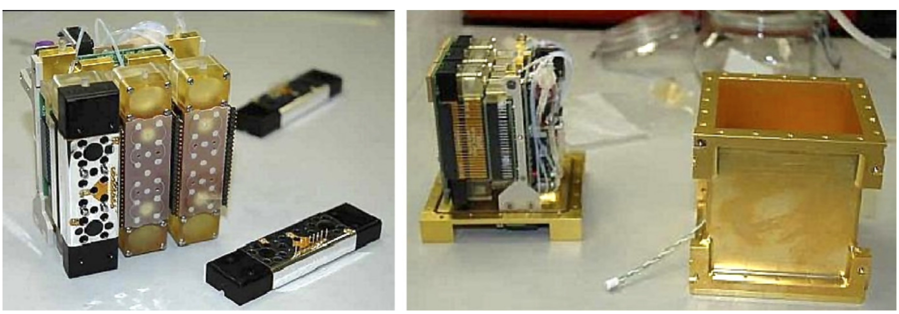

Fig. 5 SESLO's 3-bioblock payload shown partially assembled at *left* and next to its 1U hermetic container at *right*.
Credit: NASA.

4 Upcoming missions

BioSentinel is an upcoming nanosatellite mission and the first biological CubeSat that will fly into interplanetary space. *BioSentinel* is a 6U CubeSat that will fly as a secondary payload on the Space Launch System (SLS) Artemis 1 mission (formerly known as Exploration Mission-1 [EM-1]), currently scheduled to launch no earlier than 2021 [15,24–27]. A microgravity control experiment is also scheduled to fly to the ISS. The primary objective of *BioSentinel* is to investigate the DNA damage response to deep-space radiation in a eukaryotic organism, the budding yeast *S. cerevisiae*. Budding yeast was selected not only because of its known biology and flight heritage but also because of its ability to remain in desiccated state for long periods of time. Since *BioSentinel* is a secondary payload on Artemis 1, the payload has to be delivered up to 9 months prior to launch without power and/or thermal control, and the cells must also remain viable for its 6–12-month spaceflight mission, making it a requirement to keep them viable for a total of 15–21 months.

In the 6U *BioSentinel*, the biological samples and microfluidic delivery system are contained within a 4U hermetically sealed container. The 4U biosensor unit also has mounted on one of its outer surfaces a linear energy transfer (LET) spectrometer for radiation measurements and characterization. The remaining volume of the spacecraft contains all the bus components necessary for autonomous operations in deep space, including deployable, gimbaled triple-junction solar panels to generate up to 70 W; a guidance, navigation, and control unit, including three orthogonal momentum wheels, star tracker, multiple sun sensors, and an inertial measurement unit; a cold-gas micropropulsion system for detumbling upon deployment and desaturating momentum wheels; Li-ion batteries and an electric power management system; and an X-band radio/transponder (Iris, flown by the twin *MarCO* CubeSats) connected to multiple patch antennas to communicate with NASA's Deep Space Network (DSN).

Prior to payload integration, two different yeast strains will be loaded in fluidic card microwells using a trehalose stasis buffer and then allowed to air-dry to improve desiccation tolerance [26]. Growth medium and the metabolic indicator dye alamarBlue will be mixed during flight and delivered through a complex system of pumps and valves. To maintain a dry environment and thus cell viability prior to rehydration, the payload also carries silica gel-filled desiccation chambers integrated within the fluidic cards and manifold. In contrast to previous missions, the fluidic cards, Fig. 6, are made primarily of polycarbonate, so that they can be sterilized by autoclaving to prevent potential off-gassing of toxic volatiles from chemical sterilization. Each *BioSentinel* ground or flight unit will carry 18 microfluidic cards with 16 microwells each, for a total of 288 wells per unit (Fig. 6). In addition to a dedicated thermal control system for each card, each microwell is monitored using a three-color LED detection system (570, 630, and 850 nm) to take optical measurements during cell growth. After CubeSat deployment and initial system checkout, two fluidic cards will be activated by temperature increase, followed by growth medium delivery. Optical measurements will be performed using a similar system to those developed for *PharmaSat* and also used by *EcAMSat* and *O/OREOS*-SESLO (see in the preceding text).

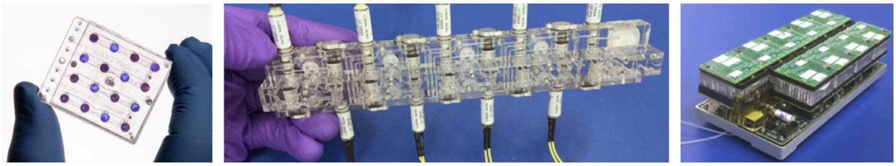

Fig. 6 *BioSentinel*'s key fluidic components. *Left*: one of eighteen 16-microwell fluidic cards. *Center*: a 9-card manifold prior to integration with fluidic cards. *Right*: a 9-card manifold integrating fluidic cards, manifold, mechanical structure, thermal control, and optical measurement subsystems. Two such fully integrated manifolds along with a growth medium and indicator dye fluidic delivery system, as well as supporting electronics, comprise the *BioSentinel* biology payload.
Credit: NASA.

The alamarBlue dye and synthetic complete (SC) growth medium will be mixed at a 1:9 ratio during flight; both the concentrated 10 × alamarBlue dye and the mixed dye-and-medium solution will be monitored using independent optical calibration cells. Cell growth and metabolic activity will be monitored using the optical detection system, and the data will be telemetered back to Earth using NASA's DSN. The biological data will be correlated to the onboard LET spectrometer data, which will provide a "space weather report" every few hours. A delayed-synchronous ground control unit carrying yeast cells from the same flight cultures will run in parallel to the Artemis 1 mission, as will another set aboard ISS.

Other future missions may be based on the preliminary work already performed on payloads such as the 2U *SporeSat-2*, designed to measure differences on *C. richardii* fern spore Ca^{2+} ion channel response and that includes 50-mm microcentrifuges with 32 ion-specific [Ca^{2+}] electrode pairs. Other ground-based work includes the GraviSat project, a 2U payload with a microcentrifuge to culture algae with pulse amplitude-modulated fluorescence and carbonate ion measurements (qualified to TRL 5) [20], and MisST, a 2U payload integrating a multistrain *Caenorhabditis elegans* fluidic habitat with a two-color fluorescence microscope (qualified to TRL 6). Now that CubeSats such as the Mars Cube One (*MarCO*) twins used during *InSight* mission to Mars have demonstrated the capability to support deep-space missions, future microbial and astrobiology CubeSats may venture farther into our solar system.

5 Discussion

To date, five microbiology or astrobiology missions have taken place in CubeSats, all developed by NASA. The success of these missions has verified the capability of this spacecraft platform to perform biological and astrobiological research. Furthermore, these free-flying satellites can provide less gravitational noise than inhabited spacecraft and stations, products of the operation of support machinery and human presence. NASA's evolutionary approach of basing a CubeSat's bus and payload on the previous mission has helped meet scientific success criteria while reducing risk and cost. Additionally, four of these missions achieved mission success thanks to technological developments that have enabled, for example, temperature control of $\pm 0.3°C$, autonomous experiment activation and termination, incorporation of centrifuges, and data acquisition and transmission to Earth. One of the key approaches contributing to the success of these missions has been the use of microfluidic cards and lab-on-a-chip devices, which in turn are becoming ever more sophisticated and powerful in terms of scientific output. The small size and mass of the CubeSat platform make it ideal for research beyond low Earth orbit, for example, to cislunar space for radiation investigations or data acquisition from geysers of Jupiter's and Saturn's moons via flybys.

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